

Laboratório de Ingestão de Dados: Sqoop

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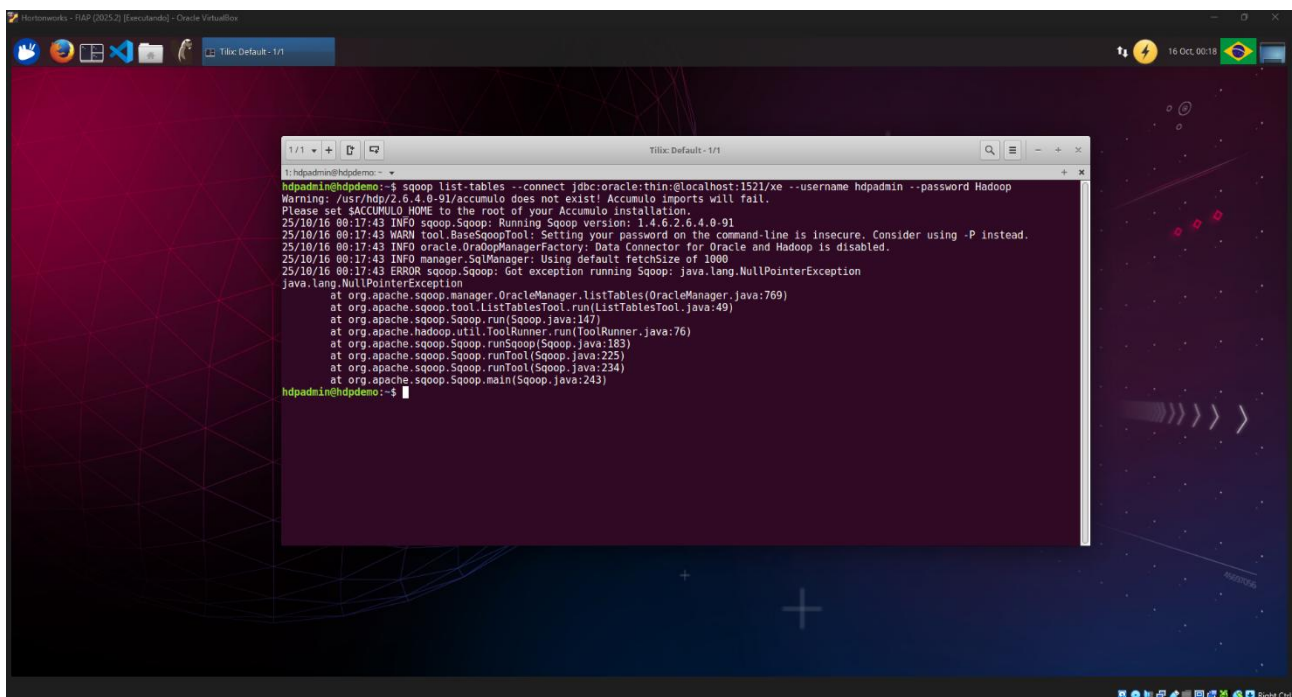
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Objetivo: Neste exercício vamos importar tabelas do banco de dados Oracle para dentro do HDFS. Posteriormente, iremos exportar dados do HDFS para uma tabela no Oracle.

Demonstração em sala: Importação das tabelas através do sqoop

1. Utilizando o Sqoop na linha de comando, liste as tabelas do database xe do Oracle:
`sqoop list-tables --connect jdbc:oracle:thin:@localhost:1521/xe --username hdpadmin --password Hadoop Hadoop`



```
hdpadmin@hdpdemo:~$ sqoop list-tables --connect jdbc:oracle:thin:@localhost:1521/xe --username hdpadmin --password Hadoop
Warning: /usr/hdp/2.6.4.0-91/accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
25/10/16 08:17:43 INFO sqoop.Sqoop: Running Sqoop version: 1.4.6.2.6.4.0-91
25/10/16 08:17:43 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
25/10/16 08:17:43 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
25/10/16 08:17:43 INFO manager.SqlManager: Using default fetchSize of 1000
25/10/16 08:17:43 ERROR sqoop.Sqoop: Got exception running Sqoop: java.lang.NullPointerException
java.lang.NullPointerException
    at org.apache.sqoop.manager.OracleManager.listTables(OracleManager.java:769)
    at org.apache.sqoop.tool.ListTablesTool.run(ListTablesTool.java:49)
    at org.apache.sqoop.Sqoop.run(Sqoop.java:147)
    at org.apache.hadoop.util.ToolRunner.run(ToolRunner.java:76)
    at org.apache.sqoop.Sqoop.runSqoop(Sqoop.java:183)
    at org.apache.sqoop.Sqoop.runTool(Sqoop.java:225)
    at org.apache.sqoop.Sqoop.runTool(Sqoop.java:234)
    at org.apache.sqoop.Sqoop.main(Sqoop.java:243)
hdpadmin@hdpdemo:~$
```

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2. Realize o import da tabela LOC_CLI_JURIDICA para o HDFS:
`sqoop import --connect jdbc:oracle:thin:@localhost:1521/xe --table HDPADMIN.LOC_CLI_JURIDICA --username hdpadmin --password hadoop`

[illegible]

3. Verifique se os dados da tabela foram importados com sucesso localizando os arquivos no HDFS.
hdfs dfs -ls /user/hdpadmin/HDPADMIN.LOC CLI JURIDICA

[illegible]



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4. Quantos mappers você acredita que foram utilizados para executar a inserção?

4 Mappers

25/10/16 00:18:44 INFO mapreduce.JobSubmitter: number of splits:4

```
hdppadmin@hdpdemo:~$ hdfs dfs -ls /user/hdppadmin/HDPADMIN.LOC_CLI_JURIDICA
```

Found 5 items

```
-rw-r--r--  3 hdpadmin hdfs      0 2025-10-16 00:18
/user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA/_SUCCESS
-rw-r--r--  3 hdpadmin hdfs    985 2025-10-16 00:18
/user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA/part-m-00000
-rw-r--r--  3 hdpadmin hdfs    957 2025-10-16 00:18
/user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA/part-m-00001
-rw-r--r--  3 hdpadmin hdfs    956 2025-10-16 00:18
/user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA/part-m-00002
-rw-r--r--  3 hdpadmin hdfs   1062 2025-10-16 00:18
/user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA/part-m-00003
```

5. Rode o comando de inserção novamente e analise o resultado. O que aconteceu?

[illegible]

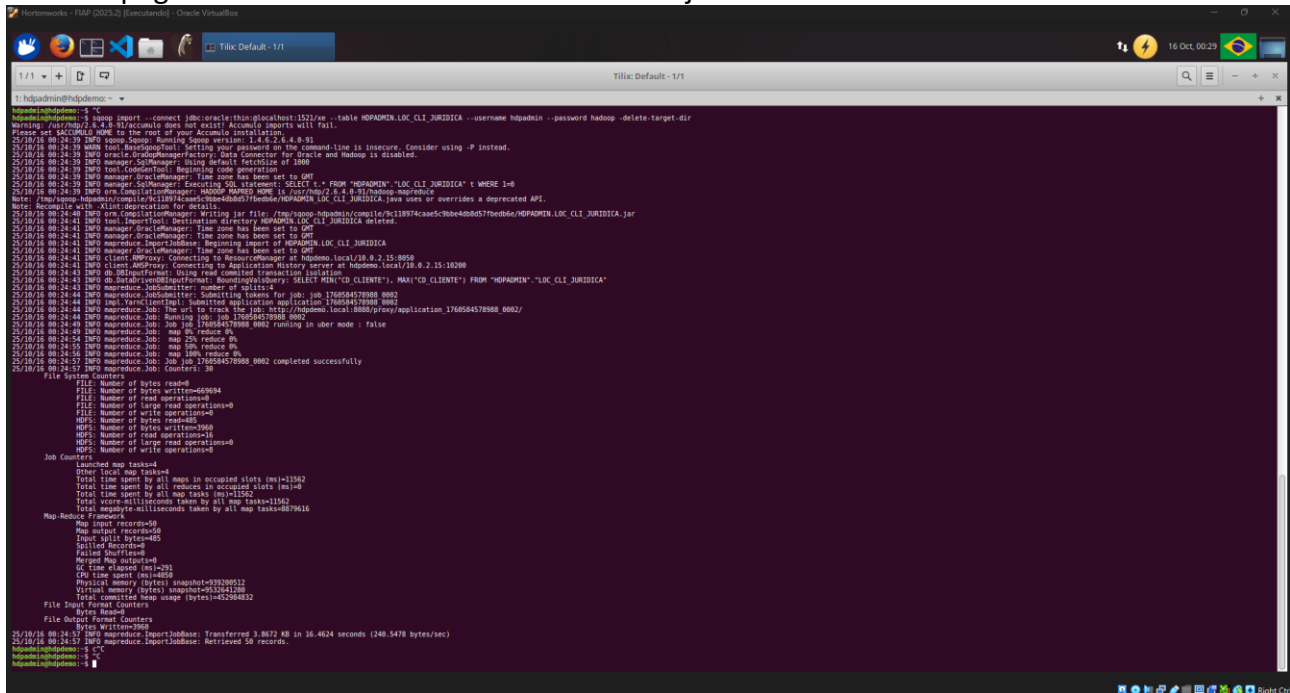
Ja existia e barrou

```
ERROR tool.ImportTool: Encountered IOException running import job:
org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory
hdfs://hdpdemo.local:8020/user/hdpadmin/HDPADMIN.LOC CLI JURIDICA already exists
```

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6. Rode o comando novamente adicionando a opção `--delete-target-dir`. Qual o resultado?

Apagou o diretório e executou novamente o job



```
hadoop@hdpdemo:~$ %  
hadoop@hdpdemo:~$ hbase import --connect jdbc:oracle:thin:@localhost:1521/hr --table HDPADMIN.LOC_CLI_JURIDICA --username hdpadmin --password hadoop --delete-target-dir  
Warning: /usr/lib64/libc.so.6: libc.so.6: version GLIBC_2.27 not found (required by /usr/lib64/libhadoop.so.1)  
Please refer to the error message for details.  
25/10/18 00:24:39 INFO SourceConn: Running Sqoop version: 1.4.6.2.6.4.0.91  
25/10/18 00:24:39 WARN ToolBridgesImpl: Setting your path based on the command-line is insecure. Consider using -P instead.  
25/10/18 00:24:39 INFO manager.ShellManager: Using default fetchSize of 1000  
25/10/18 00:24:39 INFO ToolContextImpl: Beginning code generation  
25/10/18 00:24:39 INFO manager.ShellManager: Time zone has been set to GMT  
25/10/18 00:24:39 INFO manager.ShellManager: Executing SQL statement: SELECT * FROM "HDPADMIN"."LOC_CLI_JURIDICA" t WHERE 1=0  
25/10/18 00:24:39 INFO org.apache.hadoop.mapreduce.Mapper: HADOOP MapReduce HOME is /usr/lib64/6.4.0.31/hadoop-mapreduce  
Note: Assembly with Xlint:deprecation for details.  
25/10/18 00:24:40 INFO org.apache.hadoop.mapreduce.Mapper: Writing jar file: /tmp/sqoop-hdpadmin/compile96118974/aa553b0e4ad577b0d8e/HDPADMIN.LOC_CLI_JURIDICA.jar  
25/10/18 00:24:41 INFO ToolImportImpl: Destination directory HDPADMIN.LOC_CLI_JURIDICA deleted.  
25/10/18 00:24:41 INFO manager.ImportJobSetup: Beginning import of HDPADMIN.LOC_CLI_JURIDICA  
25/10/18 00:24:41 INFO manager.ImportJobSetup: Time zone has been set to GMT  
25/10/18 00:24:41 INFO manager.ImportJobSetup: Beginning import of HDPADMIN.LOC_CLI_JURIDICA  
25/10/18 00:24:41 INFO manager.ImportJobSetup: Time zone has been set to GMT  
25/10/18 00:24:41 INFO Client: Connecting to ResourceManager at hdpdemo.local/10.0.2.15:8050  
25/10/18 00:24:41 INFO Client: Connecting to ApplicationHistory server at hdpdemo.local/10.0.2.15:10200  
25/10/18 00:24:41 INFO org.apache.hadoop.mapreduce.Mapper: Submitting job: SELECT * FROM "HDPADMIN"."LOC_CLI_JURIDICA"  
25/10/18 00:24:41 INFO mapreduce.JobSubmitter: number of splits:4  
25/10/18 00:24:41 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1760584578988_0002  
25/10/18 00:24:41 INFO mapreduce.JobSubmitter: Submitted application application_1760584578988_0002  
25/10/18 00:24:41 INFO mapreduce.Job: The url to track the job: http://hdpdemo.local:8080/progress/application_1760584578988_0002/  
25/10/18 00:24:41 INFO mapreduce.Job: Running job: job_1760584578988_0002  
25/10/18 00:24:41 INFO mapreduce.Job: Job job_1760584578988_0002 running in uber mode : false  
25/10/18 00:24:41 INFO mapreduce.Job: map 0% reduce 0%  
25/10/18 00:24:41 INFO mapreduce.Job: map 25% reduce 0%  
25/10/18 00:24:41 INFO mapreduce.Job: map 50% reduce 0%  
25/10/18 00:24:41 INFO mapreduce.Job: map 75% reduce 0%  
25/10/18 00:24:41 INFO mapreduce.Job: map 100% reduce 0%  
25/10/18 00:24:41 INFO mapreduce.Job: Job job_1760584578988_0002 completed successfully  
25/10/18 00:24:41 INFO mapreduce.Job: Counter: 30  
File System Counters  
FILE: Number of bytes read=0  
FILE: Number of bytes written=608064  
FILE: Number of read operations=0  
FILE: Number of write operations=0  
HDFS: Number of bytes read=0  
HDFS: Number of bytes written=3960  
HDFS: Number of read operations=0  
HDFS: Number of write operations=0  
Job Counters  
Launched map tasks=4  
Succeeded map tasks=4  
Total time spent by all maps in occupied slots (ms)=11562  
Total time spent by all reduces in occupied slots (ms)=0  
Total time spent by all map tasks (ms)=11562  
Total vcore-milliseconds taken by all map tasks=11562  
Total megabyte-milliseconds taken by all map tasks=8879616  
Map-Reduce Framework  
Map input records=50  
Map output records=50  
Input split bytes=485  
Spilled Records=0  
Failed Map inputs=0  
Failed Map outputs=0  
CPU time spent (ms)=4050  
Physical memory (bytes) snapshot=922304112  
Virtual memory (bytes) snapshot=961284128  
Total committed heap usage (bytes)=42260432  
File Input Format Counters  
Bytes Read=0  
File Output Format Counters  
Bytes Written=3960  
25/10/18 00:24:57 INFO mapreduce.ImportJobBase: Transferred 3.8072 MB in 16.4624 seconds (240.5476 bytes/sec)  
25/10/18 00:24:57 INFO mapreduce.ImportJobBase: Retrieved 50 records.  
hadoop@hdpdemo:~$ %  
hadoop@hdpdemo:~$ %  
hadoop@hdpdemo:~$ %
```



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Demonstração: Exportação das tabelas através do sqoop

7. Entre no cliente oracle (usuário: hdpadmin, senha:hadoop)
Sqlplus

The image shows a Windows desktop environment. At the top, there is a taskbar with several application icons: a globe, a folder, a terminal icon, and a file explorer icon. The desktop background is dark. In the foreground, a TiliX terminal window is open, displaying a MySQL database setup script. The script is titled "TiliX Default - 1/1" and is being executed in a terminal window titled "TiliX Default - 1/1". The script content is as follows:

```
1 CREATE TABLE LOC_CLI JURIDICA EXPORT (  
2   ID_CLIENTE          NUMBER(8)      NOT NULL,  
3   NO_CNPJ             NUMBER(14)     NOT NULL,  
4   INS_ESTADUAL        NUMBER(18)     NOT NULL,  
5   DT_FISCAL          DATE           NULL,  
6   INS_MUNICIPAL       NUMBER(15)     NOT NULL,  
7   NM_FANTASIA         VARCHAR2(100)  NOT NULL,  
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```



Laboratório de Ingestão de Dados: Sqoop

8. Crie a tabela LOC_CLI_JURIDICA_EXPORT no oracle:

```
CREATE TABLE LOC_CLI_JURIDICA_EXPORT (
  CD_CLIENTE      NUMBER(8)      NOT NULL,
  NR_CNPJ         NUMBER(14)     NOT NULL,
  INS_ESTADUAL    NUMBER(18)     NULL,
  DT_FUNDACAO     DATE           NULL,
  INS_MUNICIPAL   NUMBER(15)     NOT NULL,
  NM_FANTASIA     VARCHAR2(20)   NOT NULL
);
```

```
1: hdpadmin@hdpdemo: ~
SQL>
SQL>
SQL> CREATE TABLE LOC_CLI_JURIDICA EXPORT (
2  ID_CLIENTE NUMBER(8) NOT NULL,
3  NO_CNP NUMBER(14) NOT NULL,
4  INS_ESTADUAL NUMBER(18) NULL,
5  DT_FUNDACAO DATE NULL,
6  CNP_MUNICIPAL NUMBER(15) NOT NULL,
7  NM_FANTASIA VARCHAR2(30) NOT NULL
8 )
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL> SELECT table_name FROM user.tables;
TABLE NAME
-----
HP SALES
LOC_CLI_FISICA_EXPORT
LOC_CLI_JURIDICA_EXPORT
LOC VEICULO
LOC RESERVA
LOC IMPOSTARIO
LOC PREPITO
LOC RESERVA LOCAÇÃO
LOC OPERADORA DOB
LOC OPERADORA CARTAO
LOC OPERADORA
TABLE NAME
-----
LOC NO LOCAÇÃO ITEM
LOC NF LOCAÇÃO
LOC NOTAS TRANCAÇÃO
LOC NOTAS COMISSÃO
LOC NOTAS CONSUMO
LOC ITEM LOCAÇÃO
LOC IMPRIMTO
LOC GRUPO
LOC FALCENATO
LOC NOTAS VEICULO
LOC ESTAO
TABLE NAME
-----
LOC ENDERECO PROPRIETARIO
LOC ENDERECO CLIENTE
LOC DUPLICATA
LOC CLASS
LOC NOT PREPITO
LOC EDO
LOC COND PARCELAS
LOC COND PAGTO
LOC CLIENTE
LOC_CLI_JURIDICA
TABLE NAME
-----
LOC_CLI_FISICA
LOC AGENCIA
LOC CENAL
LOC CPO
LOC AGENCIA BANCO
LOC BANCO
39 rows selected.
SQL>
```

Laboratório de Ingestão de Dados: Sqoop

9. Exporte o diretório criado na primeira parte do exercício para a tabela criada:
- ```
sqoop export --connect jdbc:oracle:thin:@localhost:1521/xe --table LOC_CLI_JURIDICA_EXPORT --export-dir /user/hdpadmin/HDPADMIN.LOC_CLI_JURIDICA --username hdpadmin --password hadoop
```

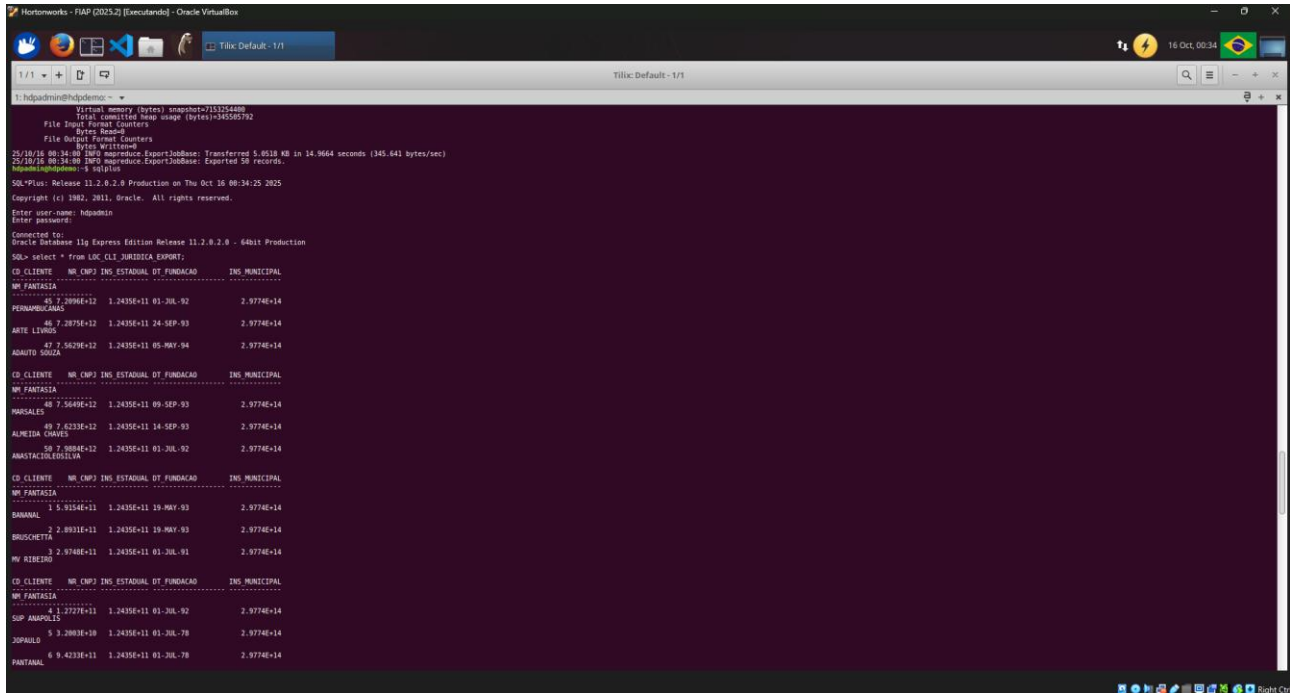
[illegible]



## Laboratório de Ingestão de Dados: Sqoop

10. Entre novamente no sqlplus e verifique que a tabela foi corretamente populada:

```
SELECT * FROM LOC_CLI_JURIDICA_EXPORT;
```



```
hdpadmin@hdpdemo: ~
11:47AM Memory (bytes) 4096000/151124480
Total committed heap usage (bytes) 345505792
File Export Export Counters
Bytes Read=
File Export Export Counters
Bytes Written=
25/10/16 09:34:09 INFO mapreduce.ExportJobBase: Transferred 5.8518 MB in 14.9664 seconds (345.641 bytes/sec)
25/10/16 09:34:09 INFO mapreduce.ExportJobBase: Exported 58 records.
HadoopJobStatus=SUCCEEDED
SQL*Plus: Release 11.2.0.2.0 Production on Thu Oct 16 09:34:25 2025
Copyright (c) 1982, 2011, Oracle. All rights reserved.
Enter user-name: hdpadmin
Enter password:
Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production
SQL> select * from LOC_CLI_JURIDICA_EXPORT;
CD_CLIENTE NR_CMP INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL

NM FANTASIA

49 7.20904E+12 1.2435E+11 01-JUL-92 2.9774E+14
PERNAMBUCAUS

46 7.2875E+12 1.2435E+11 24-SEP-93 2.9774E+14
ARTE LIVROS

47 7.5629E+12 1.2435E+11 05-MAY-94 2.9774E+14
ADALTO SOUZA

CD_CLIENTE NR_CMP INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL

NM FANTASIA

48 7.5649E+12 1.2435E+11 09-SEP-93 2.9774E+14
PIRASALES

49 7.6233E+12 1.2435E+11 14-SEP-93 2.9774E+14
ALMEIDA CHAVES

49 7.8894E+12 1.2435E+11 01-JUL-92 2.9774E+14
ANASTACIO DOS SANTOS

CD_CLIENTE NR_CMP INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL

NM FANTASIA

5 5.9354E+11 1.2435E+11 19-MAY-93 2.9774E+14
DANANAL

2 2.8931E+11 1.2435E+11 19-MAY-93 2.9774E+14
BRUSCHETTA

3 2.9748E+11 1.2435E+11 01-JUL-93 2.9774E+14
RIV REISIRO

CD_CLIENTE NR_CMP INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL

NM FANTASIA

4 1.2727E+11 1.2435E+11 01-JUL-92 2.9774E+14
SUP ANAPOLIS

5 3.2883E+10 1.2435E+11 01-JUL-79 2.9774E+14
JOPAULO

6 9.4233E+11 1.2435E+11 01-JUL-78 2.9774E+14
PANTANAL
```





# Laboratório de Ingestão de Dados: Sqoop

11. Compare o numero de registros da tabela original e da recém criada:

```
SELECT COUNT(*) FROM LOC_CLI_JURIDICA;
```

```
SELECT COUNT(*) FROM LOC_CLI_JURIDICA_EXPORT;
```

The screenshot shows a Windows desktop with a taskbar at the top containing icons for various applications. The main window is a terminal titled "Tilix Default - 1/1". The terminal displays a list of companies and their registration details, followed by a SQL query and its result.

The list of companies is as follows:

| CD_CLIENTE      | NR_CNPJ       | INS_ESTADUAL                                             | DT_FUNDACAO | INS_MUNICIPAL |
|-----------------|---------------|----------------------------------------------------------|-------------|---------------|
| MP FANTASIA     | 33 4 8086E+12 | 1.2435E+11                                               | 01-JUL-92   | 2.9774E+14    |
| TRIOSEI LTDA    | 33 4 8036E+12 | 1.2435E+11                                               | 01-JUL-92   | 2.9774E+14    |
| NACTONAL LTDA   | 33 4 8576E+12 | 1.2435E+11                                               | 12-APR-93   | 2.9774E+14    |
| CONCILAN        |               |                                                          |             |               |
| CD_CLIENTE      | NR_CNPJ       | INS_ESTADUAL <th>DT_FUNDACAO</th> <th>INS_MUNICIPAL</th> | DT_FUNDACAO | INS_MUNICIPAL |
| MP FANTASIA     | 34 4 9322E+12 | 1.2435E+11                                               | 01-JUL-92   | 2.9774E+14    |
| VENDONORTE      | 35 5.2733E+12 | 1.2435E+11                                               | 01-AUG-65   | 2.9774E+14    |
| ELDEFORSO       | 36 5.3614E+12 | 1.2435E+11                                               | 01-SEP-66   | 2.9774E+14    |
| VERISSIMO       |               |                                                          |             |               |
| CD_CLIENTE      | NR_CNPJ       | INS_ESTADUAL <th>DT_FUNDACAO</th> <th>INS_MUNICIPAL</th> | DT_FUNDACAO | INS_MUNICIPAL |
| MP FANTASIA     | 37 5.4358E+12 | 1.2435E+11                                               | 01-DEC-92   | 2.9774E+14    |
| BARAGRO         | 38 5.5212E+12 | 1.2435E+11                                               | 01-JUL-92   | 2.9774E+14    |
| MEIDA MARQUES   | 39 5.5895E+12 | 1.2435E+11                                               | 29-MAR-94   | 2.9774E+14    |
| ALBERTINO REIS  |               |                                                          |             |               |
| CD_CLIENTE      | NR_CNPJ       | INS_ESTADUAL <th>DT_FUNDACAO</th> <th>INS_MUNICIPAL</th> | DT_FUNDACAO | INS_MUNICIPAL |
| MP FANTASIA     | 40 5.8172E+12 | 1.2435E+11                                               | 01-MAY-85   | 2.9774E+14    |
| FRANDEBASTO     | 41 6.0623E+12 | 1.2435E+11                                               | 01-JUL-92   | 2.9774E+14    |
| P. RABELO E CIA | 42 6.1827E+12 | 1.2435E+11                                               | 24-MAR-94   | 2.9774E+14    |
| HAZER LEITE     |               |                                                          |             |               |
| CD_CLIENTE      | NR_CNPJ       | INS_ESTADUAL <th>DT_FUNDACAO</th> <th>INS_MUNICIPAL</th> | DT_FUNDACAO | INS_MUNICIPAL |
| MP FANTASIA     | 43 6.4673E+12 | 1.2435E+11                                               | 01-APR-89   | 2.9774E+14    |
| E. RIBEIRO LTDA | 44 6.7893E+12 | 1.2435E+11                                               | 01-AUG-78   | 2.9774E+14    |
| CANUTO LTDA     |               |                                                          |             |               |

50 rows selected.

SQL> SELECT COUNT(\*) FROM LOC\_CLI\_JURIDICA;

COUNT(\*)

50

SQL>

The screenshot shows a Windows 10 desktop with several application icons in the taskbar. The main window is Oracle VM VirtualBox, displaying a Linux virtual machine named 'Tilix: Default - 1/1'. Inside the VM, a terminal window is open, showing the output of a command that lists companies and their registration details. The output is as follows:

```

32 4.8834E+12 1.2435E+11 01-JUL-92 2.9774E+14
NACIONAL LTDA
33 4.8579E+12 1.2435E+11 12-APR-93 2.9774E+14
CONCILAR

CD_CLIENTE NR_CNPJ INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL
NM_FANTASIA

34 4.8322E+12 1.2435E+11 01-JUL-92 2.9774E+14
MINIONORTE
35 5.2733E+12 1.2435E+11 01-AUG-65 2.9774E+14
ILDEFONSO
36 5.3614E+12 1.2435E+11 01-SEP-66 2.9774E+14
W VERISSIMO

CD_CLIENTE NR_CNPJ INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL
NM_FANTASIA

37 5.4359E+12 1.2435E+11 01-DEC-92 2.9774E+14
SANGAO
38 5.5212E+12 1.2435E+11 01-JUL-92 2.9774E+14
MEIDA MARQUESE
39 5.5895E+12 1.2435E+11 29-MAR-94 2.9774E+14
ALBERTINO REIS

CD_CLIENTE NR_CNPJ INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL
NM_FANTASIA

40 5.6172E+12 1.2435E+11 01-MAY-85 2.9774E+14
FRANCOGASTO
41 6.0652E+12 1.2435E+11 01-JUL-92 2.9774E+14
P. RAULOS R CIA
42 6.1827E+12 1.2435E+11 24-MAR-94 2.9774E+14
BAZER LEITE

CD_CLIENTE NR_CNPJ INS_ESTADUAL DT_FUNDACAO INS_MUNICIPAL
NM_FANTASIA

43 6.6873E+12 1.2435E+11 01-APR-89 2.9774E+14
E. RIBEIRO LTDA
44 6.7893E+12 1.2435E+11 01-AUG-78 2.9774E+14
CANUTO LTDA

```

Below the list, the terminal shows the results of two SQL queries:

```

58 rows selected.

SQL> SELECT COUNT(*) FROM LOC_CLI_JURIDICA;
COUNT(*)

50

SQL> SELECT COUNT(*) FROM LOC_CLI_JURIDICA_EXPORT;
COUNT(*)

50

```



## Laboratório de Ingestão de Dados: Sqoop

### Desafio:

Faça a importação da tabela LOC\_CLI\_JURIDICA novamente de maneira a criar automaticamente a tabela no Hive e verifique que a tabela foi criada com sucesso. Qual o caminho dos arquivos da tabela importada no HDFS?

```
insert into loc_cli_fisica values (9999,9, 99999999999, '24-FEB-90', 999999,9,'X');
```

### Resposta

```
sqoop import \
--connect jdbc:oracle:thin:@//localhost:1521/xe \
--table HDPADMIN.LOC_CLI_JURIDICA \
--username hdpadmin -P \
--num-mappers 1 \
--fields-terminated-by ',' \
--lines-terminated-by '\n' \
--hive-import \
--hive-table loc_cli_juridica \
--create-hive-table \
--delete-target-dir
```

```
hdfs dfs -ls /apps/hive/warehouse/loc_cli_juridica
```

```
-rwxrwxrwx 3 hdpadmin hadoop 3960 2025-10-16 00:42
/apps/hive/warehouse/loc_cli_juridica/part-m-00000
```

```
SQL> SELECT *
FROM loc_cli_fisica
WHERE cd_cliente = 9999;
 2 3
CD_CLIENTE NR_CPF NR_HABILITACAO DT_NASCIMENTO NR_RG TIT_ELEITOR D

 9999 9 99999999999 24-FEB-90 999999 9 X
```



## Laboratório de Ingestão de Dados: Sqoop

[illegible][illegible]



## Laboratório de Ingestão de Dados: Sqoop

```

hdpadmin@hdpdemo ~ %
20/10/16 00:41:58 INFO mapreduce.JobSubmitter: Using read committed transaction isolation
20/10/16 00:41:59 INFO mapreduce.JobSubmitter: number of splits=1
20/10/16 00:41:59 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1760504570980_0004
20/10/16 00:41:51 INFO impl.YarnClientImpl: Submitted application application_1760504570980_0004
20/10/16 00:41:51 INFO mapreduce.Job: The url to track the job is: http://hadoop-local:8080/job/application_1760504570980_0004/
20/10/16 00:41:51 INFO mapreduce.Job: Running job: job_1760504570980_0004
20/10/16 00:41:56 INFO mapreduce.Job: Job ID: job_1760504570980_0004 running in user mode : false
20/10/16 00:41:56 INFO mapreduce.Job: map q reduce 0%
20/10/16 00:41:59 INFO mapreduce.Job: map 100% reduce 0%
20/10/16 00:41:59 INFO mapreduce.Job: Job ID: job_1760504570980_0004 completed successfully
20/10/16 00:41:00 INFO mapreduce.Job: Country: 30

File system counters
FILE: Number of bytes read=0
FILE: Bytes written=167209
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
MAPS: Number of bytes read=0
MAPS: Number of bytes written=3960
MAPS: Number of read operations=0
MAPS: Number of large read operations=0
MAPS: Number of write operations=0

Job Counters
Launched map tasks=1
Total time spent by all maps in occupied slots (ms)=1674
Total time spent by all maps in occupied slots (min)=0
Total time spent by all map tasks (ms)=1674
Total time spent by all map tasks (min)=0
Total vcore-seconds taken by all map tasks=1674
Total megabyte-milliseconds taken by all map tasks=2285632

MapReduce Framework
Map input records=50
Map output records=0
Input split bytes=0
Spilled Records=0
Failed Shuffles=0
Percent Map success=0
GC time elapsed (ms)=51
CPU time spent (ms)=706
Physical memory (bytes) snapshot=22810306
Yarn_memory_max_bytes_snapshot=2287913226
Yarn_container_map_usage(bytes)=19613376

File Input Format Counters
File Read
File Output Format Counters
Bytes Written=3960
20/10/16 00:41:00 INFO mapreduce.ImportJobBase: Transferred 3.8672 KB in 11.456 seconds (345.4705 bytes/sec)
20/10/16 00:41:00 INFO mapreduce.ImportJobBase: Retrieved 50 records
20/10/16 00:41:00 INFO mapreduce.ImportJobBase: Publishing HdfsWriteMapper import job data to Listeners
20/10/16 00:41:00 INFO mapreduce.ShuffleWriter: Time zone has been set to HIVE
20/10/16 00:41:00 INFO mapreduce.ShuffleWriter: Executing SQL statement: SELECT * FROM "HDPADMIN"."LOC_CL_JURIDICA" t WHERE t=0
20/10/16 00:41:00 INFO hive.TableParameter: Column C3 LUNATE had to be cast to a less precise type In Hive
20/10/16 00:41:00 INFO hive.TableParameter: Column MD_CNPJ had to be cast to a less precise type In Hive
20/10/16 00:41:00 INFO hive.TableParameter: Column INC_FUNDADA had to be cast to a less precise type In Hive
20/10/16 00:41:00 INFO hive.TableParameter: Column INC_FUNDADA had to be cast to a less precise type In Hive
20/10/16 00:41:00 INFO hive.TableParameter: Column INC_MUNICIPAL had to be cast to a less precise type In Hive
20/10/16 00:41:00 INFO hive.TableParameter: Column INC_MUNICIPAL had to be cast to a less precise type In Hive
Logging initialized using configuration in jar:file:/usr/hdp/2.6.4.0-91/hive/lib/hive-common-1.2.3000.2.6.4.0-91.jar!/hive-log4j.properties
0%
Time taken: 1.54 seconds
Loading data to table default_loc_cli_juridica
Table default_loc_cli_juridica Stats: (numFiles=1, numRows=0, totalSize=3960, rawDataSize=)
0%
Time taken: 0.87 seconds
hdpadmin@hdpdemo ~ % SHOW TABLES;
SHOW command not supported
hdpadmin@hdpdemo ~ % DESCRIBE FORMATTED loc_cli_juridica;
DESCRIBE FORMATTED: not supported
hdpadmin@hdpdemo ~ % --wfs a linka "location"
hdpadmin@hdpdemo ~ %
hdpadmin@hdpdemo ~ % hdfs dfs -ls /apps/hive/warehouse/loc_cli_juridica
Found 2 files.
-hwxrwxr-x 3 hdpadmin hmapsp 3960 2025-10-16 00:42 /apps/hive/warehouse/loc_cli_juridica/part-0-00000
hdpadmin@hdpdemo ~ % cat
```

```
t: hdpadmin@hdpdemo ~ -
CDFS: Number of large read operations=0
HFS: Number of write operations=2

Job Counters
Launched map tasks=1
Other local map tasks=1
Total time spent by all maps in occupied slots (ms)=1674
Total time spent by all reducers in occupied slots (ms)=0
Total time spent by all map tasks (ms)=1674
Total vcore-seconds taken by all map tasks=1674
Total megabyte-milliseconds taken by all map tasks=1285632

Map-Reduce Framework
Map input records=0
Map output records=0
Input split bytes=0
Spilled Records=0
Failed HadoopMapOutputRecords=0
Merged Map outputs=0
CG Time elapsed (msec)=0
CPU time spent (ms)=790
Physical memory (bytes) snapshot=22493536
Virtual memory (bytes) snapshot=237813216
Total committed heap usage (bytes)=13831376

File Output Format Counters
Bytes Read=0

File Output Format Counters
Bytes Written=0
21/10/16 00:41:00 INFO HiveJobDriver: Transforming 3 MB/3 KB in 11.456 seconds (343.6795 jobs/sec)
21/10/16 00:41:00 INFO warehouse.LoopJobDriver: Retrieving 50 records.
21/10/16 00:41:00 INFO HiveJobDriver: Publishing RowMetrics inputs: Job data to Listeners
21/10/16 00:41:00 INFO HDFSManager: drackonmanager: Time zone has been set to GMT
21/10/16 00:41:00 INFO HiveJobDriver: Generating SQL statements: SELECT * FROM mywarehouse "loc_cli JURIDICA" WHERE 1=0
21/10/16 00:41:00 INFO HiveTableWriter: Column C0 CLIENTE had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Column C1 CPF had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Column C2 ENDOSSEGURO had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Column C3 ESTADUAL had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Column C4 FUNDACAO had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Column C5 MUNICIPAL had to be cast to a less precise type in Hive
21/10/16 00:41:00 INFO HiveTableWriter: Loading unrelated data into Hive

Logging initialized using configuration in jar:file:/usr/hdp/2.6.4.0-9L/hive/lib/hive-common-1.2.1000.2.6.4.0-9L.jar!/hive-log4j.properties
Time taken: 1.54 seconds
Loading data to table default.loc_cli_juridica
Table default.loc_cli_juridica stats: [numFiles=1, numRows=0, totalSize=3960, rawDataSize=0]
Time taken: 0.87 seconds
hdpadmin@hdpdemo:~$ SHOW TABLES;
SHOW command not found
hdpadmin@hdpdemo:~$ DESCRIBE FORMATTED loc_cli_juridica;
DESCRIBE command not found
hdpadmin@hdpdemo:~$. ./vjeja a linha "Location"
.: command not found
hdpadmin@hdpdemo:~$ hdfs dfs -ls /apps/hive/warehouse/loc_cli_juridica
Found 1 files
-rw-rw-r-- 1 hdpadmin hadoop 3960 2025-10-16 00:42 /apps/hive/warehouse/loc_cli_juridica/part-a-00000
hdpadmin@hdpdemo:~$ sqlplus
SQL>Plus: Release 11.2.0.2.0 Production on The Oct 16 00:45:16 2025
Copyright (c) 1982..2011, Oracle. All rights reserved.
Enter user-name: hdpadmin
Enter password:
Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production
SQL> SELECT *
FROM loc_cli_fisica
WHERE cd_cliente = 9999;
CD_CLIENTE NÚ COP Nº INSCRIÇÃO DE NASCIMENTO Nº RG TIT ELECION D

9999 9 9999999999 24-FEB-96 9999999 9 X
SQL>
```